

Novel applications of Descemet's membrane for retinal diseases

Background: Descemet's membrane (DM) is the specialized extracellular matrix (ECM) located between the posterior corneal stroma and the endothelium. Thanks to its composition and mechanical properties, this basement membrane provides the biochemical cues that sustain endothelium viability and function, making it an optimal cell niche for the cell types that organize in polarized monolayers, like retinal pigment epithelium (RPE). Several decellularization techniques have been developed to isolate the ECM removing all cellular components from a variety of tissue, but it is important to select the optimal protocol that maintains ECM integrity. The present work describes the tailored-made decellularization protocol to obtain an acellular scaffold from DM, and its application for the differentiation of stem cells towards RPE lineage, with the final aim to obtain an implantable RPE graft for age-related macular degeneration treatment.

Methods: DMs were mechanically stripped from donor corneas. To induce cell lysis and eliminate all cytoplasmic and nuclear material, isolated DMs were subjected to osmotic (hypotonic) shock for two hours under constant agitation. Nuclear removal and ECM integrity after treatment were verified with DNA quantification and immunofluorescence analysis. RPE cells obtained from stem cell lines following a triphasic differentiation protocol were seeded on decellularized scaffolds, and the increase of transepithelial electrical resistance (TEER) and RPE marker expression was evaluated.

Results: DNA quantification showed a 99% reduction of total DNA in decellularized DMs compared to control ones, with no DAPI-positive nucleus still visible after decellularization. Moreover, ECM components staining maintained the same pattern and intensity as the fresh untreated control. The differentiated cells expressed typical markers of RPE phenotype, with apico-basal polarization and increased TEER reaching the threshold values corresponding to tight junction formation.

Conclusions: These results show that the decellularized DM is depleted of the donor DNA, avoiding the risk of unwanted immunogenicity when used as heterologous graft. Cells seeded on this scaffold complete their maturation towards RPE phenotype, creating a functional transplantable epithelium. DM is currently employed in intraocular surgery and prestripped by eye banks, therefore a decellularized DM could represent a valid scaffold for regenerative medicine purposes in ocular surgery.

Keywords: Age-related macular degeneration, retinal pigment epithelium, stem cells, Descemet's membrane, decellularization